

AGILE BUSINESS REQUIREMENTS DOCUMENT

SHOP-A-LYTICS

AGILE BUSINESS REQUIREMENTS DOCUMENT TEMPLATE

ITBA
Prepared for the Shop-A-Lytics Product Team

VERSION 1.0

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VERSION HISTORY				
VERSION	APPROVED BY	REVISION DATE	DESCRIPTION OF CHANGE	AUTHOR
1.0	Pending	28/09/2026	Initial draft, compiled following Client Meeting 1 and the Shop-A-Lytics concept brief	ITBA (Development Team)

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1. EXECUTIVE SUMMARY SNAPSHOT

This Business Requirements Document (BRD) sets out the functional and non-functional requirements for Shop-A-Lytics, a premium information service that turns anonymised BoschCard student transaction data into retail intelligence for local Stellenbosch retailers and FMCG brands. It draws on the Shop-A-Lytics product team's concept brief and the outcomes of Client Meeting 1 (22 September 2026), and reflects the feasibility assessment, the Must/Should/Could priorities agreed at that meeting, and the data and technical constraints the development team has since identified.

The purpose of this BRD is to give the Shop-A-Lytics product team (Kayla, Tendekai, Carel and Tinashé) and the development team, a shared, agreed reference, therefore, what will be built for the MVP release, what has been deliberately left out of scope, and which data and privacy constraints shape the design. It is written for the client-side stakeholders and the developers who will build the Shop-A-Lytics platform.

2. PROJECT DESCRIPTION

Shop-A-Lytics is a premium BoschCard information service for FMCG brands and local retailers serving the student market in Stellenbosch. It transforms anonymised BoschCard transaction data into retail intelligence, including demand trends, transaction volumes, promotion insights and anomaly detection, so that retailers and brands can make informed decisions about stock, promotions and product offerings. At present, BoschCard transaction data exists only as raw payment records. Retailers and brands therefore have no way to see spending patterns, measure the effect of their promotions, or notice unusual activity. This project transforms BoschCard's existing data into a subscription analytics product, creating a new revenue stream for BoschCard while giving retailers and brands actionable insight into student spending behaviour that they presently lack.

3. PROJECT SCOPE

The scope set out below reflects the Must/Should/Could priorities agreed with the client, within a timeline of five development sessions and roughly four build weeks. The MVP is deliberately narrow, consisting of two user roles focused on retailers, a core analytics dashboard, and an anomaly detection feature. Billing, real-time data, and the external, non-Stellenbosch, market are excluded from this build, not because they are unimportant, but because the timeline does not accommodate them.

IN SCOPE

The following are in scope for this project:

- Web-based analytics dashboard with time-based drill-down (month → week → day) and default plus custom visualisation types (e.g. bar, scatter)
- Core metrics, such as, location, foot traffic, price, discount amount, and demand changes over time, with university context (semester, exam week, recess, other key dates)
- Two MVP user roles: Admin/Owner (full drill-down) and Member (reporting and viewing only), account sign-up and login (no billing logic required)
- Report export and a categorised user feedback mechanism (in which promotion insights vs. website/system usability can be logged)
- Anomaly detection with explainable, 'plain-language' flags for unusual transaction or demand activity

OUT OF SCOPE

The following are out of scope for this project:

- Billing and subscription logic (only a placeholder for future banking-details features is included)
- Real-time data processing
- The external market beyond the Stellenbosch University student market
- FMCG-brand and product-level insights, and stock-level recommendations (not supported by the current data model, which has no product, brand, stock, or basket data)
- *Discount-to-retailer (local promotion) linking to BoschCard → to be approached tentatively and reviewed as the project progresses (as promotions aren't identifiable in the data at all right now, and discount value is being used as a rough proxy)*

4. BUSINESS DRIVERS

BoschCard is initiating Shop-A-Lytics for the following reasons:

- Provides a new revenue stream by packaging existing BoschCard transaction data as a subscription information product
- Retailer and FMCG-brand demand for visibility into student spending, category demand, and the effect of promotions
- Competitive advantage from an anomaly detection and decision support layer that goes beyond raw transaction reporting
- Ability to compare promotion periods against historical baselines, as well as provide explainable and responsible anomaly alerts

BUSINESS DRIVER #1

Shop-A-Lytics packages BoschCard's existing anonymised transaction data into a premium information product that can be sold to FMCG brands and local retailers, opening a new, incremental revenue stream from data BoschCard already holds.

BUSINESS DRIVER #2

Retailers and FMCG brands serving the student market currently have no visibility into how students spend, which categories are growing, or how their promotions affect demand. Shop-A-Lytics answers this directly with a market analytics dashboard covering demand trends, transaction volumes, and promotion insights.

BUSINESS DRIVER #3

An explainable anomaly detection layer, one that flags unusual transaction spikes or concentrated activity in plain language rather than an unexplained score, which differentiates Shop-A-Lytics from a plain reporting tool and gives BoschCard a defensible, responsible use selling point.

5. CURRENT PROCESS

BoschCard currently processes student transactions purely for payment purposes. Transaction and subscription data are captured and stored, with subscription data held separately from BoschCard transaction data, but no retailer or brand reporting exists on top of it. Retailers and FMCG brands have no way to see foot traffic, category demand, or the effect of their own promotions on student spending, and BoschCard itself has no way to identify or flag unusual transaction activity. What insight exists today comes from manual, ad-hoc requests rather than a dedicated platform.

6. PROPOSED PROCESS

Shop-A-Lytics introduces a web-based subscription dashboard that sits on top of BoschCard's existing transaction data. Anonymised and appropriately aggregated transaction data flows into a market analytics dashboard with time-based drill-down from month to week to day. Retailers and, in later phases, FMCG brands log in with role-based access, Admin/Owner or Member, to see demand trends, foot traffic, price and discount activity, and university calendar context such as semester, exam week, and recess. Promotion periods are compared against historical baselines to surface associations, not to claim causation, and an anomaly detection layer flags unusual activity with a plain-language explanation of the contributing factors rather than a raw score, leaving business users to investigate flagged activity themselves.

7. FUNCTIONAL REQUIREMENTS

The functional requirements below describe what Shop-A-Lytics must do for its MVP release. Each requirement is prioritised using the scale that follows and is mapped against the Requirements Categories (RC) table below.

PRIORITY

Value	Rating	Description
1	Critical	The requirement is critical to the project's success; without it, the project is not viable.
2	High	The requirement matters to the project's success, but the project could still be delivered as a minimum viable product (MVP) scenario.
3	Medium	The requirement adds real value to the project, but the project could still be delivered as an MVP scenario.
4	Low	The requirement is low priority, a nice-to-have, but the project's success does not depend on it.
5	Future	The requirement falls outside the project's scope and is noted here as a possible component of a future release or feature.

REQUIREMENTS CATEGORIES (RC)

ID	Requirement	Priority
RC 1	Analytics dashboard with time-based drill-down (month → week → day) and trend views	1
RC 2	Core metrics: location, foot traffic, price, discount amount, and demand changes over time	1
RC 3	University context integrated into dashboards (semester, exam week, recess, other key dates)	2
RC 4	Report export	1
RC 5	Categorised user feedback mechanism (promotion insights vs. website/system usability)	1
RC 6	Role-based dashboards: Admin/Owner (full drill-down) and Member (reporting only)	1
RC 7	Anomaly detection with explainable, 'plain-language' alerts (contributing factors, not a raw score)	2
RC 8	Promotion insights comparing activity against relevant historical baselines (association, not causation)	2

RC 9	Search bar to drill into location and other specifics	3
RC 10	Discount-to-retailer (local promotion) linking to BoschCard will be approached tentatively and reviewed as the project progresses	4

8. NON-FUNCTIONAL REQUIREMENTS

ID	Requirement
NFR 1	The dashboard must be accessible from a standard browser, with no client-side installation required.
NFR 2	Individual students must never be identifiable. Only anonymised, appropriately aggregated transaction data may be used.
NFR 3	Minimum group-size / aggregation thresholds must be applied where necessary to reduce re-identification risk.
NFR 4	Role-based access control (RBAC) must restrict each user to their own screens (Admin/Owner vs. Member) and must protect competitor confidentiality between retailers/brands.
NFR 5	Audit logs must record access to business information to support the role-based access controls.
NFR 6	Anomalies must be presented as unusual activity requiring investigation, never as confirmed fraud or account misuse.
NFR 7	Promotion and demand insights must describe what was observed in the data (association) and must not claim a promotion caused a change in sales (causation).
NFR 8	The system must remain usable and produce meaningful output even where the real (professor supplied) dataset does not match the patterns the dashboards were designed and tested against.
NFR 9	The subscription/account data model must be kept separate from BoschCard transaction data, to be consistent with the client's data separation requirement.

9. FINANCIAL STATEMENTS

The MVP build carries no billing or payment processing logic, so it has no direct revenue impact during the build phase. Its value lies instead in validating a subscription information product, that can generate future licensing revenue from FMCG brands and local retailers once a billing capability is added in a later release.

10. COST AND BENEFIT

Costs for the MVP are limited to development effort, the five client sessions plus roughly four build weeks of developer time covering data modelling, the dashboard build, and anomaly detection logic, along with web hosting and database infrastructure for testing and the demo build. No specialised hardware is required at this stage. All tools used are free or open source, so no licensing costs are expected at MVP stage. The benefit is a working, demonstrable subscription analytics product, one that can be extended with billing and a broader FMCG-brand feature set in a future release.

11. RESOURCES

PERSONNEL

- Client-side:
Product Manager (Kayla), Quality Assessor (Tendekai), UI Lead (Carel), and Data Manager (Tinashe).
- Development-side:
ITBA (Nkazimulo), Project Manager (Iman), Lead Developer (Christiaan), QA Lead (Bourgeoise).

SOFTWARE AND PLATFORM

Need	Proposed resource
Collaboration, version control, task tracking	Notion
Data model and design artefacts	dbdiagram.io or draw.io (ERD for the additional tables) and Figma free plan (wireframes and user flow)
Database	PostgreSQL. Subscription and account data is kept separate from BoschCard transaction data (NFR 9).
Web application and access control	Django (Python), including login, role-based access (Admin/Owner and Member), and audit logging.
Analytics and anomaly detection	pandas, SQL aggregation, and rule-based/statistical checks with plain-language explanations.
Dashboard and visualisation	Django templates, Bootstrap 5, and Plotly.
Report export	CSV export and PDF export (or print view).
Testing	Django test framework or pytest, plus a manual checklist mapped to the requirements.
Demo environment	Local deployment with a backup screen recording. A free-tier cloud host may be used if terms allow.

DATA AND CLIENT-SUPPLIED INPUTS

- A synthetic BoschCard transaction dataset for build and testing, designed to include both patterned and noisy variants so the system stays usable if the real data differs (NFR 8).
- The client's ERD and data dictionary, expected after Session 2. The team's own data model work covers only the additional tables (promotion periods, university calendar, feedback, audit log, and anomaly flags).
- University calendar data (semester, exam weeks, recess, other key dates) from the client.

No data source has yet been confirmed for campus events or paydays.

- Promotions are not identifiable in the current data. Discount information is therefore being investigated as a proxy for promotion insight.
- The real (professor supplied) BoschCard dataset, arriving approximately two days before the final deadline. A data loading and validation step will be prepared in advance.

INFRASTRUCTURE

Web hosting and database infrastructure are needed for testing and the demo build only. Free or local options are planned, so no infrastructure costs are expected.

12. SCHEDULE, TIMELINE, AND DEADLINES

The project runs across five weekly Tuesday client sessions plus roughly four build weeks. Session 1 (22 September 2026) covered concept confirmation, Must/Should/Could scoping, and data and technical constraints. Scope is to be frozen by Session 3 (6 October 2026). The client is to share the ERD and data dictionary at Session 2, so that ERD work can begin, and to confirm the data model diagram, wireframes, requirements, and feasibility at that same session. The real BoschCard/professor dataset is expected approximately two days before the final deadline.

13. ASSUMPTIONS

The assumptions below, factors believed to be true but not yet confirmed, shape this BRD and its risk profile.

- The real BoschCard (professor supplied) dataset will only be available approximately two days before the final deadline, thus development and testing will use synthetic mock data until then.
- Vendor types (retailer vs. FMCG brand) are set by the lecturer and may be coarse (e.g. a single broad "Retail" type), which would flatten category level comparisons.
- It is not yet confirmed whether FMCG brands have their own vendor entries (just marked with a different type_id from retailers), or whether they don't appear in the Vendor table at all and only get access to aggregated data across retailers. This decides whether role-based access can use one simple model for all vendors, or whether brand accounts need separate, aggregate-only access rules. To be confirmed with the client (see Appendix).
- Promotions are not directly identifiable in the data (no promotion ID or date range). Discount value will be used instead.
- There is no data source for exams, campus events, or paydays, therefore university context features will rely on whatever calendar data the development team can acquire.
- "Historical baseline" for comparing promotion periods is not yet defined and will be proposed by the development team and confirmed with the client (e.g. the same weekdays over previous weeks).

- Discounts are assigned at the BoschCard transaction level. Local/retailer level discount linking to BoschCard is a tentative, lower priority feature to be reviewed as the project progresses.
- The subscription entity is separate from the BoschCard transaction entity in the data model.
- Shop-A-Lytics is not limited to BoschCard users, but MVP analysis will draw primarily on BoschCard transaction history, of which approximately five years is available (based on assumption and client preference).
- Billing/subscription logic, real-time data processing, and the external (non-Stellenbosch) market are out of scope for this build and are assumed not to affect MVP feasibility.

14. GLOSSARY

The following terms and abbreviations are used throughout this document.	
Term / Abbreviation	Explanation
MVP	Minimum Viable Product, the smallest version of Shop-A-Lytics that delivers value for this build.
FMCG	Fast-Moving Consumer Goods, the brands whose products are sold through local retailers to students.

15. REFERENCES

The following client-supplied materials were referenced in preparing this BRD.	
Name	Location
Shop-A-Lytics Concept Brief	Provided by client (Kayla, Tendekai, Carel, Tinashe)
Client Meeting 1 – Meeting Notes (22 September 2026)	Compiled by development team
Client Meeting 1 – Preparation Notes	Compiled by Nkazi (ITBA)

16. APPENDIX

The items below were flagged during Client Meeting 1 as lower priority and were not resolved live. They are listed here as open follow-up questions, to be confirmed with the client by email before or during Session 2.

- Does the Vendor entity include FMCG brands as vendors in their own right, or does it represent retail outlets only (i.e. does any Transaction ever reference an FMCG-brand vendor_id, or does every transaction resolve to a retail outlet)? Is there any existing link between a retail outlet and the FMCG brand(s) it sells?
- What is the grain of a transaction record (e.g. one row per purchase, per basket, or per card swipe)?
- What would the client consider a fair "historical baseline" for comparing promotion periods (e.g. the same weekdays over the previous few weeks), given what we will propose in Session 2?
- Should anomaly flags appear as notifications, or just be readable on the dashboard?
- Are there any existing dashboards (e.g. BoschCard's own admin panel, a retail analytics tool) the client would like the team to take design cues from?